

Fine-mapped single-cell eQTL evaluation of AlphaGenome fine-tuning

1 Overview

We evaluate whether sequence-derived mutation effects distinguish fine-mapped single-cell expression quantitative trait loci (eQTLs). A positive is a variant with posterior inclusion probability (PIP) greater than 0.75 and a negative is a variant with PIP less than 0.01. The evaluation uses human fine-mapping panels and a 1 Mb sequence window centered on the transcription start site (TSS) of the associated gene.

The primary comparison is the pretrained AlphaGenome trunk with a cell-type-matched semantic RNA head versus the two best available joint fine-tuned models, a LoRA model and a LoRA+LoCon model. The benchmark retains up to 1,000 variants in each PIP class per fine-mapping panel with a fixed random seed. The raw and retained counts are recorded with the benchmark output.

2 Cell-type matching

The 35 fine-mapping panels use the prefixes Ast, End, Ext, IN, MG, OD, and OPC, with numbered panels denoting subpanels. The base AlphaGenome comparator uses a semantic-bootstrap-initialized copy of the human Zemke 2023 RNA head. The fine-tuned models retain the same target track definitions. The matching used for every model is shown below.

Table 1: Fine-mapping panel to RNA-track matching. A numbered panel inherits the mapping for its prefix. The base and fine-tuned models use the same matched RNA tracks.

Panel prefix	Base AlphaGenome	LoRA and LoCon
Ast*	ASC	ASC
End*	Endo	Endo
Ext*	Mean of L2/3 IT, L4/5 IT, L5/6 NP, L5 ET, L5 IT, L6 CT, L6 IT CAR3, and L6b	Same
IN*	Mean of LAMP5, PVALB, SNCG, SST, and VIP	Same
MG*	MGC	MGC
OD*	ODC	ODC
OPC*	OPC	OPC

This is the closest available cell-type match because the pretrained model does not provide native single-cell RNA tracks with the fine-mapping labels.

3 Mutation effects

For variant i , let $\mathbf{y}_{i,r} \in \mathbb{R}^{B \times C}$ and $\mathbf{y}_{i,a} \in \mathbb{R}^{B \times C}$ be the reference and alternate RNA predictions at the model’s 128 bp resolution, where B is the number of output bins and $C = 19$ is the number of RNA tracks. Let $\mathcal{C}_g$ be the track indices for the matched cell group. The TSS score is

$$s_i^{\text{TSS}} = \frac{1}{|\mathcal{C}_g|} \sum_{c \in \mathcal{C}_g} |y_{i,a,b_{\text{TSS}},c} - y_{i,r,b_{\text{TSS}},c}|, \quad (1)$$

where b_{TSS} is the 128 bp bin containing the TSS. The gene score is the corresponding mean absolute alternate-reference difference after averaging each track over all 128 bp bins overlapping the annotated gene span,

$$s_i^{\text{gene}} = \frac{1}{|\mathcal{C}_g|} \sum_{c \in \mathcal{C}_g} \left| \frac{1}{|\mathcal{B}_i|} \sum_{b \in \mathcal{B}_i} y_{i,a,b,c} - \frac{1}{|\mathcal{B}_i|} \sum_{b \in \mathcal{B}_i} y_{i,r,b,c} \right|, \quad (2)$$

where $\mathcal{B}_i$ is the set of bins overlapping the gene span. Both scores are nonnegative because fine-mapping identifies variant probability, not the direction of expression change. Performance is measured by AUROC and average precision, overall and within bins of $|\text{variant} - \text{TSS}|$.

4 Results

The completed benchmark is summarized by distance from the TSS in the following figures. The underlying per-bin AUROC, AUPR, and class counts remain available in the machine-readable summary output.

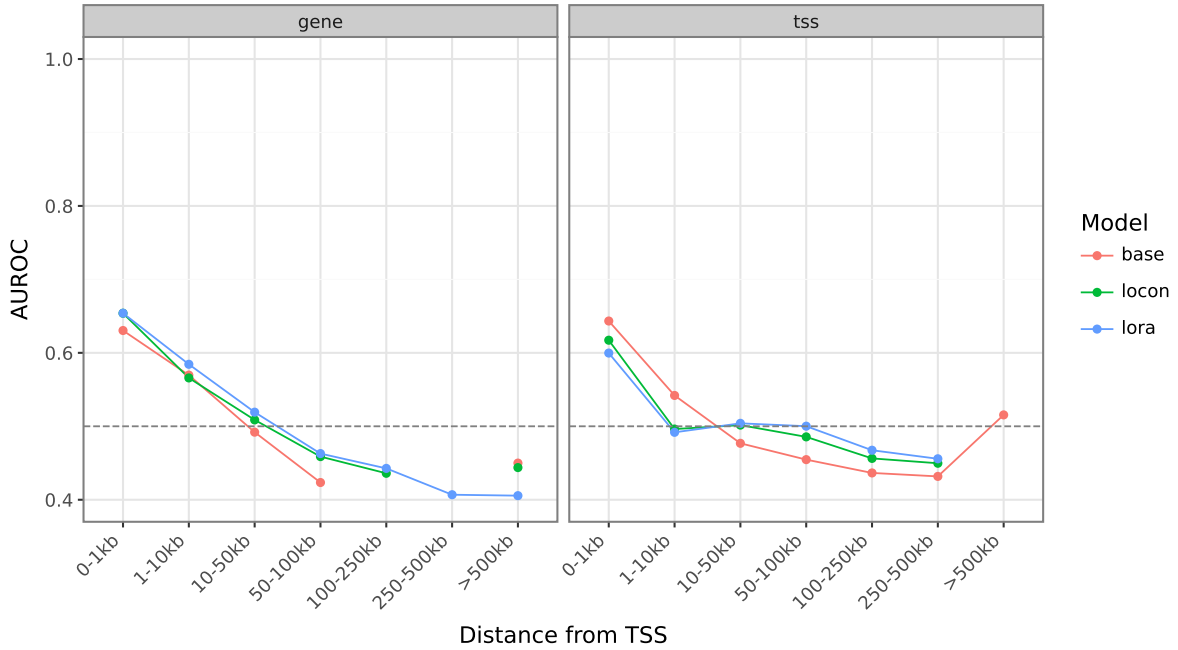


Figure 1: AUROC by distance from the TSS for TSS-bin and gene-span mutation effects. The dashed line is chance performance.

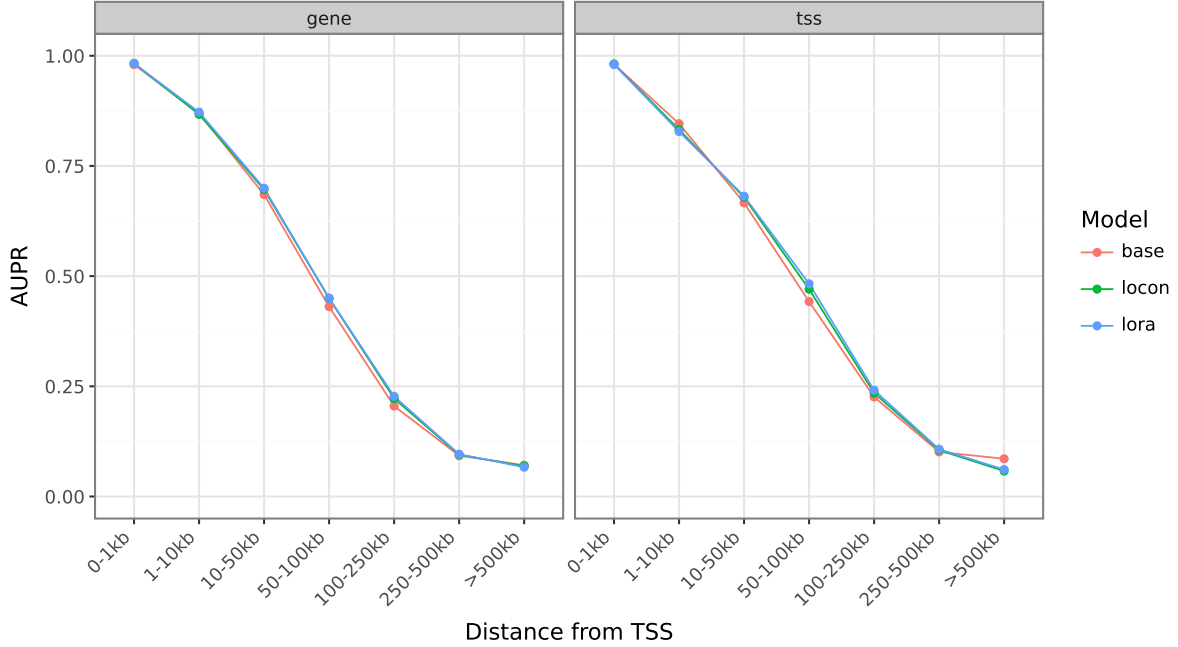


Figure 2: AUPR by distance from the TSS for TSS-bin and gene-span mutation effects.

5 Implementation details

Fine-mapping tables are streamed from compressed PIP files. Variant alleles are checked against the reference genome before constructing the alternate one-hot sequence. The same reference and alternate sequences are evaluated for every model. The base comparator uses the pretrained trunk and the same 20-track RNA head, initialized by semantic bootstrap from compatible pretrained neural tracks; it is therefore a matched-head baseline rather than an uninitialized random head. Fine-tuned models retain their trained heads and adapters. All model prediction calls use the custom forward path so the fine-tuned heads, rather than the delegated pretrained variant scorer, are evaluated.